

Supplementary Information: Accurate quantum-centric simulations of intermolecular interactions

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Supplementary Note 1: Geometries of potential energy surfaces

The PES for the water dimer is calculated for distances between oxygen atom of the first water molecule and the hydrogen atom facing this oxygen atom from the second water molecule. In our calculations the corresponding distance ranges between 1.400 and 3.500 Å. We distribute the points for water dimer PES as 1.400, 1.500, 1.600, 1.700, 1.800, 1.900, 1.962, 2.000, 2.100, 2.200, 2.300, 2.400, 2.500, 3.000, and 3.500 Å. All of water dimer simulations are done for all of these distances. The PES for the methane dimer is calculated for the distances between two carbon atoms ranging between 2.500 and 6.000 Å. We distribute the points for methane dimer PES as 2.500, 2.750, 3.000, 3.167, 3.334, 3.500, 3.667, 3.834, 4.000, 4.250, 4.500, 4.750, 5.000, and 6.000 Å. To calculate the total energy of unbound dimer we utilize the distance of 48.000 Å for both water and methane dimers. All of CASCI (16e,16o), CCSD, CCSD(T), CCSD (16e,16o), and CCSD(T) (16e,16o) simulations of methane dimer as well as SQD (16e,16o) simulations with $|\tilde{\chi}_b| = 20.0 \cdot 10^3$ are done for all of the distances described earlier. The methane dimer CASCI (16e,16o) simulations and SQD (16e,16o) simulations with $|\tilde{\chi}_b| = 20.0 \cdot 10^3$ are also performed for an additional distance of 3.638 Å. The SQD (16e,16o) energy extrapolations using $|\tilde{\chi}_b|$ of $9.0 \cdot 10^3$, $11.0 \cdot 10^3$, and $14.0 \cdot 10^3$ are done for 4.000, 4.250, 4.500, 4.750, 5.000, 6.000, and 48.000 Å distances. In the case of HCI (16e,24o) simulations of the methane dimer, we use only distances of 3.638, 3.667, 3.750, 3.834, and 3.900 Å. In case of SQD (16e,24o) simulations we only use the distance of 3.638 Å. All calculations are done as single-point energy calculations with no geometry optimizations. To produce the geometries studies in this work, we start from the equilibrium geometries and change the distance between the centers of the monomers, with the geometries of the individual monomers fixed.

Supplementary Note 2: Details of HCI calculations

In HCI simulations instead of generating all of the single and double excitations one generates only those single and double excitations that correspond to Hamiltonian matrix elements exceeding a threshold ε . In HCI ε_1 controls which determinants will be included in the variational wavefunction. In our HCI calculations, we use values of ε_1 equal to $5 \cdot 10^{-6}$ (during initial variational steps) and

$1 \cdot 10^{-6}$ (during later variational steps). We do not use the non-variational perturbative correction in our HCI calculations. Hence, our HCI calculations are fully variational, which allows for more appropriate comparison between the SQD and HCI results.

Supplementary Note 3: Mapping of Electronic Configurations with Jordan-Wigner Transformation

In this work the electronic configurations (Slater determinants) are mapped to qubit states through the Jordan-Wigner (JW) transformation as

$$|\mathbf{x}\rangle = \prod_{\rho\sigma} (\hat{a}_{\rho\sigma}^\dagger)^{x_{\rho\sigma}} |\emptyset\rangle, \quad (\text{S1})$$

where $|\emptyset\rangle$ is the vacuum state corresponding to zero electrons, $\hat{a}_{\rho\sigma}^\dagger$ is the creation operator, and $x_{\rho\sigma} \in \{0, 1\}$ is the element of the computational basis, $|\mathbf{x}\rangle$, represented as,

$$|\mathbf{x}\rangle = \bigotimes_{p\sigma} |x_{p\sigma}\rangle. \quad (\text{S2})$$

Here each bitstring corresponds to the spin-up/down orbitals of the studied system as

$$\mathbf{x} = (x_{N_{\text{MO}}-1\downarrow} \dots x_{0\downarrow} x_{N_{\text{MO}}-1\uparrow} \dots x_{0\uparrow}) \quad (\text{S3})$$

where the first half of the bitstring represents the spin-down orbitals, while the second half represents the spin-up orbitals as $\mathbf{x}_\downarrow = (x_{N_{\text{MO}}-1\downarrow} \dots x_{0\downarrow})$ and $\mathbf{x}_\uparrow = (x_{N_{\text{MO}}-1\uparrow} \dots x_{0\uparrow})$, respectively. Since the JW mapping operates in terms of spin orbitals, as oppose to molecular orbitals (MOs), the JW mapping employs $M = 2N_{\text{MO}}$ qubits. Utilization of the spin orbitals allows for the computation of the number of spin- σ electrons for a given configuration $\mathbf{x}$ as $N_{\mathbf{x}\sigma} = \sum_p x_{p\sigma}$, while the total number of electrons is calculated as $N_{\mathbf{x}} = \sum_\sigma N_{\mathbf{x}\sigma}$. The $N_{\mathbf{x}}$ corresponds to the Hamming weight of $\mathbf{x}$, where in the right particle sector the $N_{\mathbf{x}} = N_\sigma$.

Supplementary Note 4: Effect of number of samples on total energy in SQD(16e,16o)

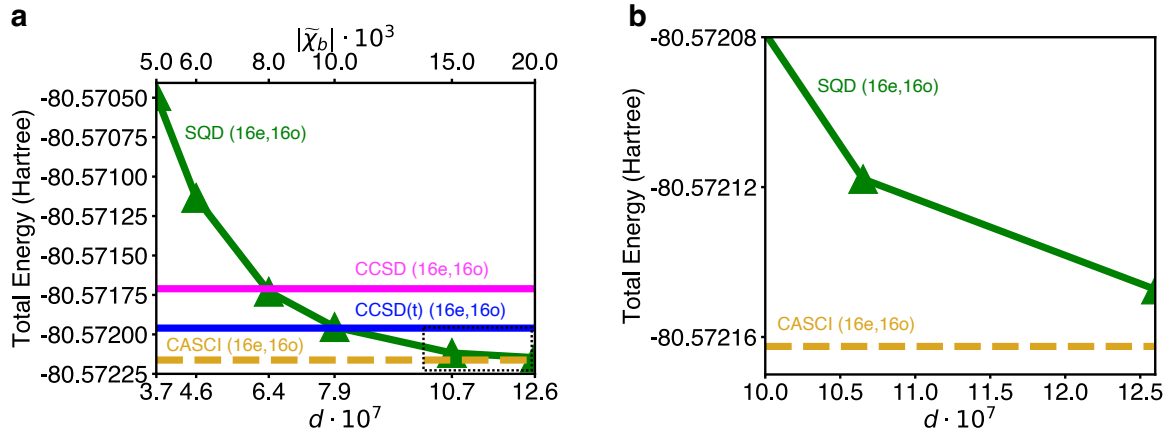


Fig. S1 Total energy of methane dimer predicted with SQD (16e,16o) at 3.638 Å distance between the monomers as the function of $d \cdot 10^7$. (a) the entire range of d , and (b) a magnified region with largest values of d , highlighted in panel (a) as a black box. The secondary x-axis demonstrates the value of $|\tilde{\chi}_b| \cdot 10^3$ producing the given value of $d \cdot 10^7$. Solid green line with triangular markers shows SQD (16e,16o) results. Horizontal magenta, blue, and dashed light brown lines indicate the total energies from CCSD (16e,16o), CCSD(T) (16e,16o), and CASCI (16e,16o) calculations.

Figure S1 demonstrates that for the methane dimer the SQD (16e,16o) calculations with $8 \cdot 10^3$ samples per batch have a total energy below the CCSD (16e,16o) value while increasing the number of samples per batch to $10 \cdot 10^3$ leads to the recovery of the total energy of CCSD(t) (16e,16o). Increasing the number of samples per batch to $20 \cdot 10^3$ results in a SQD (16e,16o) total energy that is only 0.010 kcal/mol away from the total energy of CASCI (16e,16o).

Supplementary Note 5: PES of methane dimer including the repulsive region

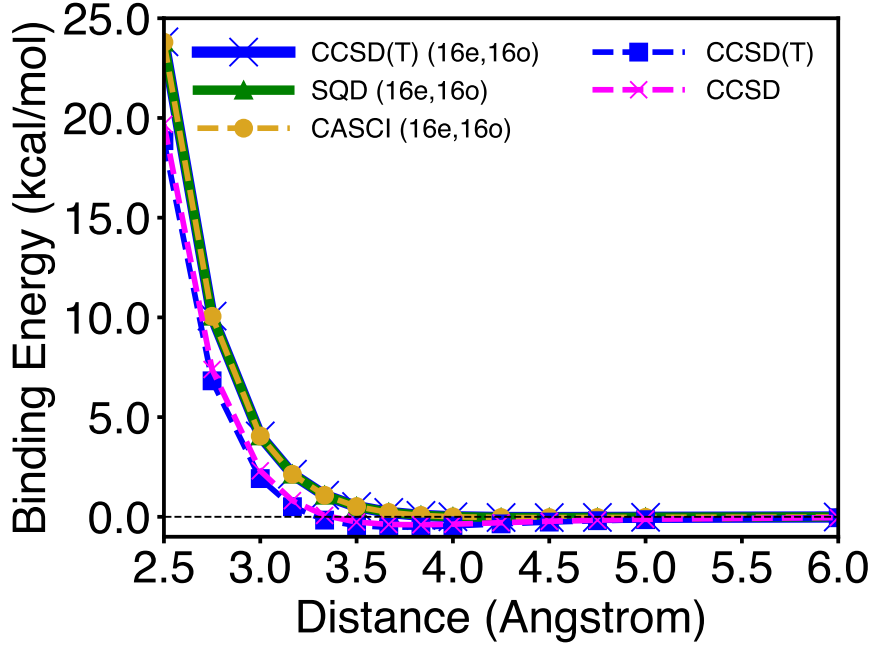


Fig. S2 Binding energies of the methane dimer along its PES, where the distances between the centers of methane molecules range between 2.500 and 6.000 Å. Active space simulations are performed with (16e,16o). Light brown, orange, and blue dashed lines with circle markers depict PES calculated with CASCI, CCSD, and CCSD(T) methods, respectively. Solid yellow line with triangular markers depicts the PES calculated with the SQR method. Solid blue line represents CCSD(T) calculations using an active space. Black horizontal dashed line indicates the zero value of the binding energy.

Supplementary Note 6: Effect of the improved LUCJ parametrization

While SQR with LUCJ parameterized using t_2 amplitudes from CCSD has shown promising results, significant efforts have also been devoted to improving the distribution obtained from quantum simulations using LUCJ circuits[1, 2]. A key approach involves optimizing the parameters through fitting based on the two-body operators to enhance energy estimation. As discussed earlier in Methods section LUCJ employs a local approximation, and its two-body operator differs substantially from that in the original unitary coupled cluster theory:

$$\hat{T}_2 - \hat{T}_2^\dagger = \sum_{abcd} \mathcal{T}_{abcd} \hat{a}_d^\dagger \hat{a}_c^\dagger \hat{a}_d \hat{a}_b \quad (\text{S4})$$

Such difference can be reduced by minimizing the following functional:

$$\chi^2(J, K) = \sum_{abcd} |\mathcal{S}_{abcd} - \mathcal{T}_{abcd}|^2, \quad (\text{S5})$$

where

$$\mathcal{S}_{abcd} = \sum_{\mu} \sum_{pqrs} [e^{K^\mu}]_{pa} [e^{-K^\mu}]_{rb} [e^{K^\mu}]_{qc} [e^{-K^\mu}]_{sd} \delta_{pr} \delta_{qs} J_{pq}^\mu. \quad (\text{S6})$$

This minimization was performed using *scipy.optimize* with the Powell method. The target functional as indicated in Eq. (S5) was reduced from 0.376 to 0.271. Figure S3 below presents the LUCJ

energy, representing the SCI calculation based solely on determinants with the correct number of particles before any configuration recovery. The figure presents the LUCJ energy of a methane dimer calculated with an active space of (16e, 16o) at an intermolecular distance of 3.638Å, plotted as a function of $|\tilde{\chi}_b| \cdot 10^3$. The solid green line with triangular markers represents LUCJ (16e, 16o) results obtained using t_2 amplitudes from coupled-cluster singles and doubles (CCSD) on the `ibm.cleveland` quantum device. In contrast, the solid yellow line with triangular markers corresponds to LUCJ (16e, 16o) energies obtained by parameter fitting, which effectively lowers the energy compared to using t_2 amplitudes alone. Furthermore, the solid brown line with circular markers represents LUCJ (16e, 16o) results from parameter fitting in noiseless simulations (using the `ffsim` library [3]), demonstrating the potential of this approach in a low-noise quantum environment. The results suggest that parameter fitting improves energy estimates on current quantum hardware, and the noiseless simulation highlights the promise of this technique for future quantum devices with reduced noise. We are actively exploring this direction to enhance quantum sampling and improve energy calculations on next-generation quantum processors.

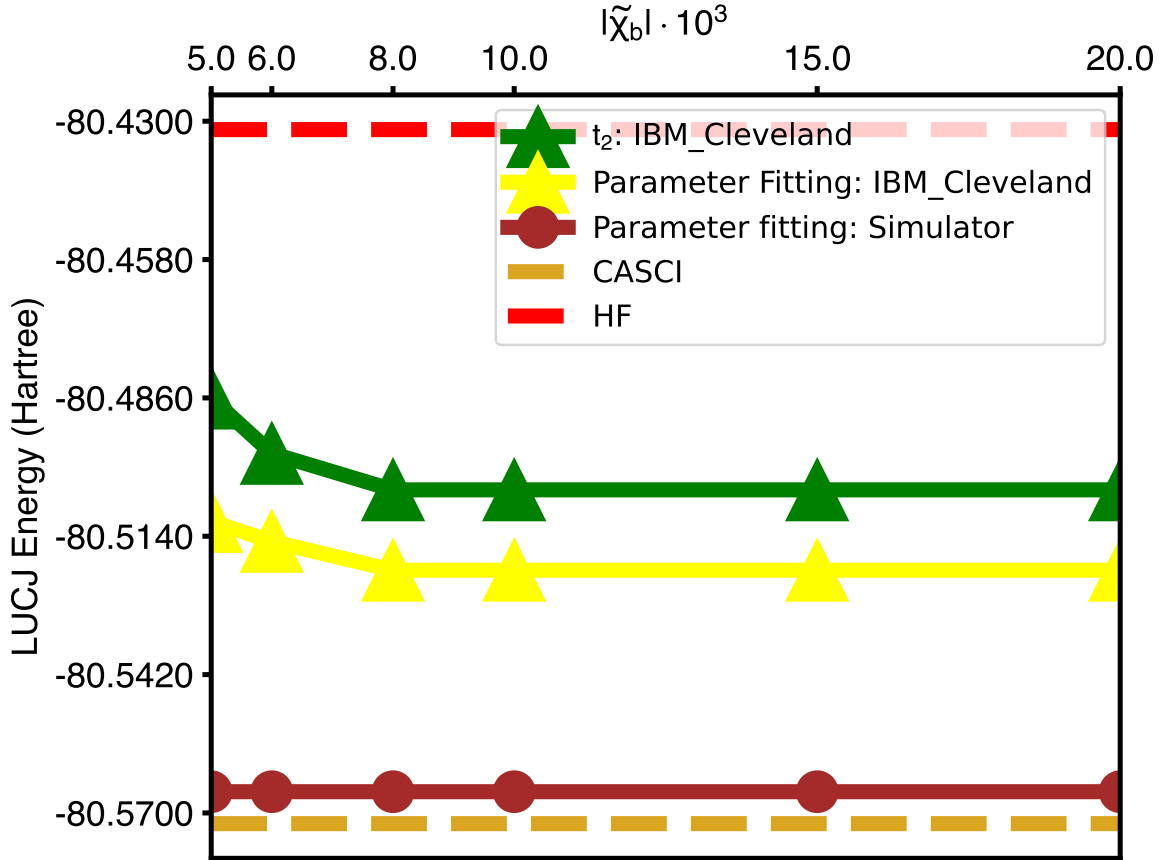


Fig. S3 LUCJ energy of a methane dimer calculated with active space (16e, 16o) at a distance of 3.638Å, plotted as a function of $|\tilde{\chi}_b| \cdot 10^3$. The solid green line with triangular markers represent LUCJ (16e, 16o) results using t_2 amplitudes from CCSD, while the solid yellow line with triangular markers corresponds to LUCJ (16e, 16o) results obtained through parameter fitting. The solid brown line with circles represent LUCJ (16e, 16o) energies with parameter fitting from noiseless simulations. The dashed red and light brown denote LUCJ energies from HF and CASCI (16e, 16o) calculations, respectively.

Supplementary Note 7: DMET SQD methodology and scalability analysis

Density Matrix Embedding Theory (DMET) [4–8] is a wavefunction-based quantum embedding approach that enables the application of high-level electronic structure methods to selected subsystems while treating the rest of the system at a mean-field level. This method effectively reduces the computational cost of beyond mean-field calculations while maintaining accuracy.

DMET partitions the D -dimensional Hilbert space of a quantum system into a fragment subspace of dimension d_F (where $d_F \ll D$) and an environment subspace of dimension $D_E = D - d_F$. The exact ground-state wavefunction of the system Hamiltonian, $\hat{H}$, can be expressed using a Schmidt decomposition:

$$|\Psi\rangle = \sum_{\mu=1}^{d_F} \lambda_{\mu} |u_{\mu}\rangle \otimes |b_{\mu}\rangle, \quad (\text{S7})$$

where the bath states are given by $|b_{\mu}\rangle = \sum_e V_{\mu e} |e\rangle$. These bath states span a space of dimension d_F , significantly reducing the complexity of the environment. The ground-state solution of $\hat{H}$ can be obtained from the embedding Hamiltonian:

$$\hat{H}_{emb} = \hat{P} \hat{H} \hat{P}, \quad (\text{S8})$$

where the projector $\hat{P} = \sum_{\mu\nu} |u_{\mu}\rangle \langle u_{\mu}| \otimes |b_{\nu}\rangle \langle b_{\nu}|$ defines a lower-dimensional subspace. Although this formulation is theoretically exact, it requires prior knowledge of the system's ground state, making direct implementation impractical.

To make DMET computationally feasible, a mean-field approximation is employed [6]. For a closed-shell system with N_{elec} electrons in N_{orb} spatial orbitals, the mean-field wavefunction is given by a Slater determinant:

$$|\Phi\rangle = \prod_{i\sigma} \hat{a}_{ci\sigma}^{\dagger} |\emptyset\rangle, \quad (\text{S9})$$

where $|\emptyset\rangle$ represents the vacuum state and orbitals $|c_i\rangle = \sum_p C_{pi} |p\rangle$ are expressed in a localized basis. The orbitals are categorized into fragment ($p \in F$) and environment ($e \notin F$) groups.

To define the bath orbitals, the environment block of the one-particle reduced density matrix is diagonalized:

$$\rho_{ee'} = \langle \Phi | \hat{E}_{e'}^e | \Phi \rangle. \quad (\text{S10})$$

The eigenvalues δ_{α} classify the orbitals into (i) virtual environment orbitals ($\delta_{\alpha} = 0$), (ii) bath orbitals ($0 < \delta_{\alpha} < 2$), and (iii) occupied environment orbitals ($\delta_{\alpha} = 2$). This transformation effectively maps the problem onto an active space representation [5, 6].

The full-system electronic Hamiltonian:

$$\hat{H} = \sum_{pr} t_{pr} \hat{E}_r^p + \sum_{pqrs} \frac{(pr|qs)}{2} \hat{E}_{rs}^{pq}, \quad (\text{S11})$$

is mapped to an effective subsystem Hamiltonian:

$$\begin{aligned} \hat{H}_{emb} &= \sum_{tu} \tilde{t}_{tu} \hat{E}_u^t + \sum_{tuvw} \frac{(tu|vw)}{2} \hat{E}_{uw}^{tv}, \\ \tilde{t}_{tu} &= t_{tu} + \sum_{ee'} \left[(tu|ee') - (te'|eu) \right] \tilde{\rho}_{ee'}, \end{aligned} \quad (\text{S12})$$

where t, u refer to fragment and bath orbitals, and the effective density matrix of occupied environment orbitals is given by $\tilde{\rho}_{ee'} = \sum_{\alpha: \delta_{\alpha}=2} W_{e\alpha} W_{e'\alpha}^*$. High-accuracy methods such as full configuration interaction (FCI) [4, 5], density matrix renormalization group (DMRG) [9], complete active space self-consistent field (CASSCF) [8, 10], and coupled cluster singles and doubles (CCSD) [11] can be applied to solve for the subsystem ground state.

When using multiple overlapping fragments in DMET, variations in electron number and energy accuracy may arise [5, 6]. To address this, we employ a “one-shot” approach [6], where the embedding Hamiltonian is modified by introducing a global chemical potential:

$$\hat{H}_{emb} \rightarrow \hat{H}_{emb} - \mu_{\text{glob}} \sum_{f \in F} \hat{a}_f^{\dagger} \hat{a}_f. \quad (\text{S13})$$

The parameter μ_{glob} is iteratively optimized to ensure that the electron number predicted by DMET matches the exact system electron number, thus improving accuracy and consistency.

Figure S4 illustrates the application of DMET-SQD in scaling the SQD approach to efficiently treat the (36e, 36o) active space of cyclohexane in its chair conformation. Here cyclohexane simulations are done for sto-3g basis set thus (36e, 36o) corresponds to FCI case. Due to the computational

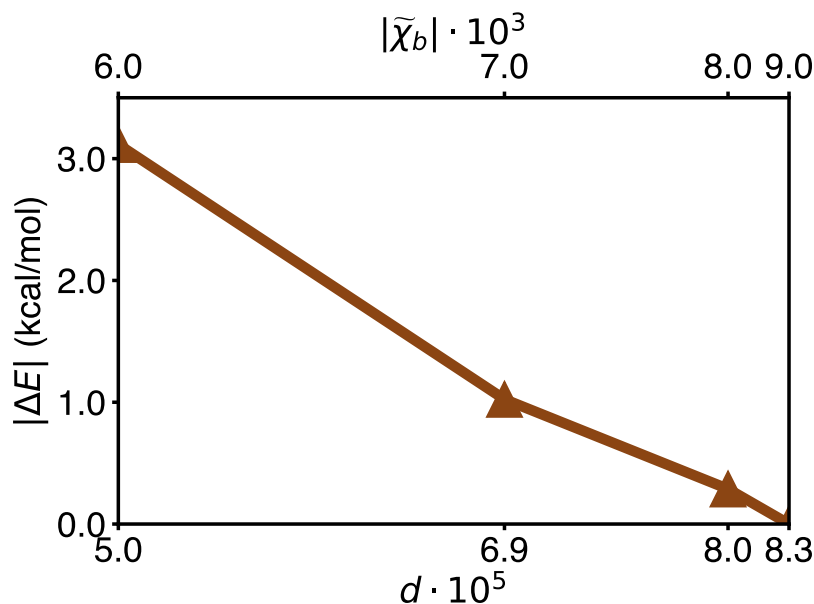


Fig. S4 Absolute energy differences ($|\Delta E|$) in kcal/mol for the chair conformer between DMET-SQD and DMET-FCI as a function of $d \cdot 10^5$. Dark brown triangular markers depict the computed deviations at each sampling point. The secondary x-axis demonstrates the value of $|\tilde{\chi}_b| \cdot 10^3$ producing the given value of $d \cdot 10^5$.

challenges associated with solving the full $(36e, 36o)$ active space, the system is fragmented into smaller subsystems, each with an active space of $(14e, 14o)$, making the calculations significantly more tractable while retaining accuracy. The geometry of cyclohexane in its chair conformation is computed with MP2/aug-cc-pVDZ level of theory with ORCA 5.0.4 software package. [12]

The plot presents the absolute energy deviations ($|\Delta E|$) between DMET-SQD and the DMET-FCI reference, revealing a systematic reduction in energy difference as the value of $|\tilde{\chi}_b| \cdot 10^3$ increases. The deviations range from approximately 3.10 kcal/mol for the sparsest sampling down to just 0.015 kcal/mol at the highest sample density. These results highlight the importance of increasing the number of samples to improve accuracy, as finer sampling enables better interpolation between data points and minimizes errors.

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